Why Time Runs Forward: Simple Laws Must Grow Minds in Causal Steps

Abstract

Time runs forward because a simple universe has to grow complex minds, and growth takes causal steps. The universe starts simple, as observation shows, and on the measure argued here because each bit of a complete description, laws and starting state together, halves its probability. Minds like ours are complex, so the universe has to grow complexity from that simple start, as many simple rules do: the Game of Life's one rule about neighbors grows five live cells into more than a thousand generations of structure and can run any computation. On the view taken here the steps are discrete, since a continuum has no next step for a cause to produce; on either view, growth runs one way. Time is the causal steps, each a rule applied to rules, so it runs from simple to complex.

The reverse fails. An endless chain of steps has a first step and no last one, so the whole history has no end to run back from. A finite reversal has to say which step it turns back from, and each bit of that address halves its weight. Minds run only forward: a thought merges many possible states into one, so run backward a brain's present state no longer fixes its next one, which would have to be set by heat converging from its surroundings. Memories are built by steps already taken. So we remember the past and not the future.

Keywords: arrow of time; causation; algorithmic probability; Kolmogorov complexity; emergence of complexity; discreteness; psychological arrow of time; mathematical monism

1. The argument

The universe starts simple. A complete description of a universe gives its laws and its starting state, and each bit of that description is a 50/50 choice that no rule makes, so each added bit halves its probability. Among universes that fit the same evidence, the simplest dominate. Observation shows the same start, whatever explains it: the laws we know fit on a page, and the early universe was smooth to about one part in a hundred thousand, with no structures yet, let alone minds (Planck Collaboration, 2020). The argument below needs the simple start as a fact, not the measure (the probability weighting just described) as its reason, and a reason can fail without taking the rest of the chain with it. Everything that exists is ruled or random, and both are mathematical, so time has to come out of that mathematical structure and from nothing outside it.

Minds are not simple. A brain holds an enormous amount of organized information. Each bit of it is either written into the description, halving the probability, or grown by the laws from a simpler state. The probable universes are the ones that grow their minds.

Many simple rules grow complexity. The Game of Life updates a grid of cells by one rule about how many neighbors are alive. Start it with five live cells, the shape called the R-pentomino, and it churns for more than a thousand generations, throwing off gliders that travel across the grid; suitable patterns in it can run any computation (Gardner, 1970; Berlekamp, Conway and Guy, 1982). What grows is organization. The description stays short, since every later state is fixed by the rule, the start and the number of steps. Physics does the same at every scale. Classical paths, the definite trajectories of non-quantum physics, emerge from the quantum sum over all paths (Feynman, 1948), and entropy rises as the steps spread energy into more arrangements.

Growth takes causal steps, and on the view taken here they are discrete. A cause produces its effect, and producing an effect takes a step from one state to the

next. A continuum has no next state: between any two instants lies a third, so no state is the first one a cause produces. That is an a priori proof in the logician's sense, and Alfred North Whitehead reached the same conclusion from Zeno's paradoxes (Whitehead, 1978). Others hold that a continuous law makes each state depend on earlier ones without a first effect, as in Russell's view that motion is only being at different places at different times (Russell, 1903). A position for every instant, with nothing that produces the next, is the emperor's new points. The argument of this paper holds on either reading: growth from a simple start runs one way, and an endless history has no last moment to run back from. A length can be measured as a continuum, and its points do not cause it.

On the same view, change without a cause is outside logic and reason: nothing could derive it from anything, explain it or predict it, and an account that allows it could not say why anything happens. Indeterministic readings of quantum theory accept such change; the direction of time argued here does not depend on rejecting them.

Time is the causal steps, and it runs forward: from simple to complex, from cause to effect. The second direction is a priori, since a cause produces its effect (Section 2); the first rests on the weighting of descriptions (Section 3). A billiard break shows the asymmetry. One instruction sends fifteen balls scattering, sound through the room and warmth into the table and the air. Reversing the break takes every ball, every air molecule and every vibration in the table returning together in perfect coordination. Written from the end, the reversal needs every particle listed. The only short way to describe it is the long way around: start from the simple beginning, run the break forward, then reverse from a chosen step.

Minds run forward too. Each thought is produced from the one before, and memories are built by steps already taken, which is why we remember the past and not the future.

Everything that happens is such a step, a rule applied to rules. On the view taken here, the rules are the only things. A structure that only sat there would still exist, as mathematics does, and do nothing and know nothing, like a dead brain with every connection intact. Time is the applying, step after step. Space is how far the applying spreads, and the speed of light is its base rate: a body's steps at that rate are shared between moving and changing in place, so a moving clock runs slow. Light, which crosses space while no time passes for it, is the limit and no exception: it spends every step moving, so nothing within it could move relative to the rest.

2. A required beginning and an optional ending

Why production needs a source

If every operation waits for an earlier operation, none has run, and no input is supplied independently, none can start. Initiated generation therefore needs supplied information that enables at least one operation. For successive states linked by one transition rule, that information can be a starting state. A simpler rule does not supply it by itself.

Processing the next event

The next event is the next thing produced. One operation's output enables the operation after it, and that chain of productions is what temporal succession consists of. Time needs no additional continuum of physically instantiated instants between those events.

The real line, the continuous number line, has no next instant: between any t<t' lies (t+t')/2. A process that must move to the immediate next real instant therefore has no such step. Counting events by stages gives a separate result. If

generation occurs at natural-number stages (counted 0, 1, 2 and so on), introducing finitely or countably many (endlessly many yet listable one by one) elementary events at each stage, the resulting event set is at most countable. A continuum model need not work in stages: x(t)=t specifies a trajectory at every real time without enumerating its instants. Such a model says where each state sits. It does not say what produces each state, and physical succession is production, so it comes in stages.

The reason production needs a next state is a regress. Suppose a cause at one instant produced states at later instants of a continuum. Every later instant has others between it and the cause, so no later state is produced by the cause directly; each would be produced by states nearer still, and the chain of production has no first link, as Zeno's runner has no first step. Whitehead, arguing by Zeno's method, put it this way: once anything becomes, there is "no continuity of becoming" (Whitehead, 1978, p. 35). A differential equation (an equation giving how fast a quantity changes) such as dx/dt = u(t) meets no such regress, because it states how each state depends on what came before without saying that anything produces anything. That is causation as dependence. Production is the stronger relation, and physical succession needs it, since a law of dependence alone says which histories fit together and not why any of them happens. The direction of time does not hang on this point. A history in continuous time that starts simple would still have to grow its minds, and an endless future would still have no last moment to run back from; what production adds is that the steps are discrete.

Why a beginning does not require an end

A supplied beginning enables the first operation; it imposes no final operation. A rule may terminate after finitely many events or continue without a last event.

Forward generation can supply event 0 and then obtain event n+1 from event n. The sequence can continue indefinitely. It needs no completed infinity, an endless series taken as already finished, before a finite event occurs.

Run the whole history backward instead and event n waits on the already-produced event n+1. With no final event, and no intermediate state supplied independently, that reverse chain has no first operation that can run. A state supplied part-way along can start a finite reversal. It cannot hand the complete endless history a last event.

Why continuation costs nothing extra

A nonhalting generator needs no further instruction to continue. Specifying a stopping point costs extra only when the generator and shared decoder (the agreed machine that reads descriptions) do not already determine it. A timestamp, threshold or trigger may supply that information; a law can also contain a short stopping rule from the outset. The comparison must count the complete programs before assigning a penalty.

Production within the block

Production here means a dependence between events within the history. It does not mean that an external machine creates an otherwise unreal block, the whole history existing at once. A rooted dependence graph, a network of what depends on what, can have a first node even though the complete graph exists timelessly.

Calling one direction forward does not by itself make it the direction experience runs. A completely reversed observer has reversed records and record-forming relations too. The symmetry breaks at the start and in the observer. The forward history is produced from a simple supplied state. A reversed history has to be produced from its complex end, written in bit by bit or obtained by running the

forward history first, and Section 4 shows that a reversed brain does not produce its own next state.

Joint specification and physical succession

A simple law need not describe a history with a beginning. The rule x_t=t for every integer t in Z describes one with no first event. If instead each state must be generated from the previous state, the process cannot start unless some state is supplied. A separate algorithm can enumerate the labels 0,1,-1,2,-2,... and compute their associated states, though that order of computation is not the history's temporal order. Computing every state in some order is not the same as producing each event from its temporal predecessor.

The premise doing the work is causality. Whatever is produced has a producer, so a chain of production traced back reaches a supplied first state instead of running on without an origin. Denying the premise means accepting events that nothing produces, and an account with such events cannot say why anything in it happens. A block universe keeps every event; the premise says how the events in it are related, as producer and produced. Appendix A takes up the objection that the complete block needs no production.

3. Simple beginnings and the cost of finite reversal

Laws and starting conditions form one account

The production argument requires a starting state; the preference for short descriptions, made precise just below, supplies a reason to favor an economical one. Count the laws and initial information together, allowing each to explain the other and counting shared information once. A short law with a separately specified elaborate boundary can cost more than a slightly longer law that

generates its boundary.

The cost of any extra specification can be counted exactly, and it compounds. Stated precisely, following Solomonoff (1964), assign a complete description of length L the weight 2^(-L), using one stated prefix-free binary language, one in which no description is the start of another. If a reverse construction keeps the forward program and adds k encoded bits, then

$$\frac{w_{\text{reverse}}}{w_{\text{forward}}} = \frac{2^{-(L+k)}}{2^{-L}} = 2^{-k}.$$

Every bit that is truly additional matters. One extra bit halves the description's weight; ten reduce it by a factor of 1,024; sixteen by 65,536. The original program's size does not dilute the penalty.

From descriptions to observer probabilities

A prior weight, the weight given before any evidence, becomes a prediction about experience only after we say how an observer is sampled and what evidence it has. In symbols, for each finite history H, let N_H be its total number of eligible observer moments and n_H(D) the number with evidence D. Their fraction is f_H(D)=n_H(D)/N_H. Assume the relevant denominators are positive and w_R=2^(-k)w_F.

Choosing a history by its prior and then sampling uniformly (each moment equally likely) within that history gives

$$\frac{P(R \mid D)}{P(F \mid D)} = 2^{-k}\frac{f_R(D)}{f_F(D)}.$$

Giving each eligible moment its history's weight instead gives

$$\frac{P(R \mid D)}{P(F \mid D)} = 2^{-k}\frac{n_R(D)}{n_F(D)} = 2^{-k}\frac{N_R f_R(D)}{N_F f_F(D)}.$$

If the matching fractions are equal, k=16 and N_R/N_F=2^20, history-first sampling favors F by 65,536 to one, while moment-first sampling favors R by sixteen to one. These are different sampling assumptions. Neither the code penalty nor the total observer count settles the odds alone. Infinite collections require a specified measure in place of uniform finite sampling.

A complete reversal changes the observer's records along with its environment. It does not automatically lower the likelihood of that observer's accessible evidence. With equal evidence likelihoods under history-first sampling, a shorter complete description retains its prior advantage; this does not force the absolutely shortest law or starting state.

The reversal rule and its trigger

The forward generator saves the cost of listing the later state particle by particle. In exchange it needs a reversal procedure and a point at which to apply it, and if those go in as extra encoded instructions, their cost is real.

Saying when to reverse carries a price that can be counted. Stated precisely, if a separate fixed-length binary field must distinguish N otherwise unspecified reversal points, it needs at least ceil(log2 N) bits. There are only 2^k different k-bit fields, so 1,024 distinct choices require ten bits (Grunwald and Vitanyi, 2008).

A compressed endpoint illustration

An arbitrary endpoint may cost much more to specify than a specially chosen one. Roughly fourteen billion years contains about 8x10^60 Planck-time positions (ticks 5.4x10^-44 seconds long), between 2^202 and 2^203; a fixed-width address for an arbitrary position therefore needs 203 bits. A rule such as "stop at a specified power of two" may specify a particular endpoint much more briefly, once the language for that rule is declared. If a complete reverse description is sixteen

bits longer in the chosen code, its assigned weight is lower by 65,536. That last statement is exact; the sixteen-bit difference itself must be established by the comparison.

The same accounting applies to an event-defined endpoint: "reverse at the first observer satisfying criterion C" must specify C and its evaluation unless the common model already supplies them.

A code can charge both directions equally, and the asymmetry stands

A finished string, a sequence of symbols, read in either order costs about the same to describe. For a finite encoded history h and a fixed computable reversal R, a program for R(h) can run a program for h and apply R:

$$K(R(h)) \leq K(h) + c_R.$$

If R is its own inverse, so that applying it twice restores the original:

$$|K(R(h)) - K(h)| \leq c_R.$$

Appendix B goes further and constructs a universal code (one able to imitate any other description language) that assigns both orientations exactly the same shortest length. These results concern a finished string, and the asymmetry defended here lies in what produces what. An endless history produced from a simple start has no last state to reverse from. A finite reversal has to locate its turning point inside a forward history, and when the history has no shortcut, the cheap programs for a reversed string get it by running the forward history first and reversing the output. A history with a shortcut, such as counting 0, 1, 2 and so on, can be reversed directly by counting down from N, and it still has to be told N. A reversed world produced from its own start would begin at the forward history's last state, which, for a history with no shortcut, has no short description except as the end of the forward run; supplied any other way, that start has to be written in

bit by bit. A reversed brain does not produce its own next state (Section 4). A code that charges both readings of a finite string equally leaves all three in place.

4. Why minds run forward

Simple laws stay the same while the structures they grow gain organization. Suppose the starting state lacks the organization observers require and a later state contains observers who have it. The history must build that organization, and if stages are indexed by the natural numbers there is a first stage at which it is present.

A brain runs forward because its thinking merges. Adding 1 and 1 gives 2, and so does adding 0 and 2. A brain's step takes its present state and its input to one next state, and many different earlier states lead to the same later one, with the difference carried off into the surroundings as heat (Landauer, 1961; Bennett, 1973). A brain runs on about 20 watts, all of it ending as heat; a computer built to keep every bit could retrace its steps, and a brain keeps no such record. Forward, a brain's next state is fixed by its present state and its input. Backward, they leave it open: $2 = x + y$ has more than one solution, and nothing in the brain's present state picks one. In a fully reversed history the choice is made by heat flowing back in from the surroundings in perfect coordination. A brain whose next state is set from outside, by converging heat, is not producing its own states, and producing each state from the last is what thinking is.

Records run the same way. A memory is a correlation between the recording system and what it recorded, built by a step already taken. Leonard Mlodinow and Todd Brun (2014) show that any memory that does not have to be fine-tuned to what it records points in the direction in which entropy rises. A reversed observer's "records" would be correlations with what its history produces next, built by nothing it did. Watch a reversed break without seeing the incoming

coordination and the reassembly looks like a sudden coincidence. For a reversed learner every regularity would look that way, since the convergences depend on details it cannot sense (Price, 1996). When two situations supply a learner with exactly the same observations and memory while successful action requires incompatible responses, no decision rule reading only that information can succeed in both. Heinrich, Knopp and Pas (2015) compare survival in a toy environment (a deliberately simplified model world) of scattering or converging rocks.

Reversing the environment while leaving the learner's recording processes alone does not reverse the complete history. A complete reversal transforms the records and the way they form, and that is where the argument above applies: the reversed brain's transitions are set by its converging surroundings and not by its own organization.

5. How entropy enters the account

Growing minds spreads heat. Organisms build local organization while dispersing heat and waste into their surroundings, so the steps that build minds are steps in which entropy rises. What remains is why the start was low in entropy, and simplicity alone does not settle that.

A simple beginning need not be a low-entropy beginning

Two states can be equally easy to describe while sitting in macrostates (sets of exact states sharing the same overall properties) of very different size. Stated precisely, consider n binary sites, with n even, and let a macrostate record only the number of ones. The all-zero state belongs to a macrostate of size one. The alternating state 0101... belongs to the largest macrostate, of size binomial(n,n/2). Both states have descriptions of length K(n)+O(1). Yet their Boltzmann entropies

under this coarse-graining (this grouping of states by their number of ones) are 0 and k_B ln binomial(n,n/2). Short individual description and low macrostate entropy are different properties.

A reversible cyclic shift, moving each bit one place around a loop, preserves the alternating state's number of ones at every step. It supplies a short law, a short starting state and no entropy increase for this macro-observable. Short descriptions alone therefore do not yield a thermodynamic arrow.

Gravity joins simple and low-entropy

In our universe the two properties coincide, because of gravity. In a gas without gravity, an even spread is the high-entropy state. Under gravity, clumping is the direction of rising entropy, so a smooth, even early universe, the simplest start to describe, is also a low-entropy one (Penrose, 2004). The cosmic microwave background, the faint radiation left over from the early universe, shows that start: the early universe was even to about one part in 100,000. The simple start that the measure favors is the low-entropy start that the second law (entropy's tendency to rise) needs.

A count of states gets part of the way there. Under uniform preparation (every starting state equally likely), the move toward the larger set of states is the more probable one. Stated precisely, consider a finite reversible system. Two macroscopic conditions A and B contain N_A and N_B states. If m states in A evolve into B, exactly m states in B return to A under inverse evolution. Uniform preparations give probabilities m/N_A and m/N_B. Their ratio is N_B/N_A. Entropy S=k_B ln N expresses the ratio as exp((S_B-S_A)/k_B). These probabilities concern the weighted preparations; they require no fundamental randomness.

The formal results in this area come with assumptions that need stating. Related fluctuation relations compare thermodynamic probabilities under specified conditions (Crooks, 1999). Janzing, Chaves and Scholkopf (2016) give an algorithmic result: under one-to-one dynamics, in which no two states merge into one, state description length cannot decrease beyond a fixed additive constant when the initial state is algorithmically independent of the dynamics (neither helps describe the other more briefly). That independence is an additional premise.

The pieces fit. The measure favors a simple start. Under gravity a simple start is a low-entropy start. The steps that grow minds from it spread energy and information into the surroundings, so entropy rises in the same direction in which minds remember.

6. Objections

"The Past Hypothesis already explains the arrow." David Albert (2000) adds to the time-symmetric laws (laws that allow every history's reverse as well) a postulate that the universe began in a state of very low entropy, and derives the arrows of entropy, records and influence from it. This account keeps that start as an observed fact (Section 1) and adds two things. It gives a reason to expect the start, on the measure argued here, since under gravity the simplest start to describe is also a low-entropy one (Section 5). It also ties the direction of minds to production, each state of a mind produced from the last (Section 4), where the Past Hypothesis ties it to records formed as entropy rises. The two accounts agree on the direction. If the measure fails, they stand on the same observed start, and the account of minds holds on either.

"The laws are symmetric in time, so every forward history has a reversed twin." Josef Loschmidt's objection holds against any derivation of the arrow from time-symmetric laws alone, since for each history those laws allow, they allow its

reverse. This account derives the arrow from a start and from production, so it grants the twin. The twin fits the laws, and as a finished string it costs about as much to describe as the forward history (Section 3). What it lacks is a producer. Produced from its own start, it would begin at the forward history's end, which for a history with no shortcut has to be written in bit by bit (Section 3), and its brains would have each next state set by heat converging from their surroundings (Section 4). Symmetric laws make the two histories equally lawful, and only one of them is produced from its start.

"The direction is a feature of our perspective." Huw Price (1996) argues that the laws themselves give time no direction, and that the asymmetry of cause and effect reflects the standpoint of agents who deliberate and act. The account here agrees about the laws. It locates the asymmetry in two features of the history itself: a simple supplied start, and a rooted chain of production running from it (Section 2). Minds take their direction from that chain, and the merging steps of Section 4 make it one-way with no appeal to agents, so the perspective Price describes follows from the account and is not among its premises.

"Physics has no place for causes." Bertrand Russell (1913) argued that the laws of physics state functional dependence, equations linking quantities to one another, and that the notion of cause plays no part in them. Read that way, this paper's chain of production is a gloss on dependence. The direction of time argued here survives the reading. A history that starts simple, under laws of dependence alone, still has to grow its minds; an endless future still has no last moment to run back from (Section 2); and a brain's next state is fixed by its present state and its input only in the forward direction (Section 4). What the reading removes is the discreteness of the steps and the line Appendix A draws between production and dependence, both held here as views; the direction needs neither.

"Description length is not physics." Algorithmic probability began as a tool for prediction, and no law of physics weighs histories by the length of their descriptions. The direction of time argued here does not need one to. It needs the simple start, which observation shows (Section 1), and the growth of minds from it in steps (Sections 2 and 4). The measure gives a reason to expect that start, and a reader who rejects the measure keeps the fact and the rest of the argument.

"Time could run both ways from a simple middle." Julian Barbour, Tim Koslowski and Flavio Mercati (2014) find that in a gravitating system of many particles, a measure of complexity is smallest at one point of each typical history and grows in both directions away from it, so the history has one past and two futures, and observers on either side find time running away from that point. The account here allows the same. A simple state can sit in the middle of a history, production can run outward from it both ways, and the minds on each side are built by steps taken away from it, so each side remembers toward the middle. Neither side reverses the other, since neither runs back from a complex end.

"The past could be endless." Some cosmologies extend time back without limit, and an endless past has no first state to supply. The account here needs a supplied state and does not need it to come first: a supplied state can sit anywhere in a history and root its production, and minds grow away from it, as in the case above. A history in which each state depends on an earlier one and none is supplied is a block of dependence, and Appendix A takes up why a block of that kind holds no running minds. Inflation, a very rapid expansion of the early universe, does not by itself give an endless past, since a spacetime that has expanded on average along a path traced back into the past cannot be extended along it without limit (Borde, Guth and Vilenkin, 2003).

7. Conclusion

Time runs forward because a simple universe has to grow complex minds, and growth takes causal steps. The universe starts simple, as observation shows, and on the measure argued here because each bit of its description halves its probability. Minds are complex, so the probable universes grow them, as simple rules grow gliders and computers in the Game of Life. Growth is causation, discrete on the view taken here, since a continuum has no next state; on either view it runs from the simple start. Time is the chain of those steps, each a rule applied to rules, running from the simple start toward the minds it grows.

The reverse has nowhere to begin. An endless chain has no last step to run back from. A finite reversal pays for the step it turns back from, and its cheap descriptions run the forward history first. A reversed brain's next state is set by heat converging from its surroundings and not by its own present state. A code that charges both readings of a finite string equally changes none of this, because the asymmetry lies in what produces what.

Gravity makes the simple start a low-entropy one, and the steps that grow minds spread heat into their surroundings. Memories are built by steps already taken. Minds remember the past and not the future because the past is where they were built.

Appendix A. Production, blocks and memory

The argument's premise is causality: whatever is produced has a producer. Trace back what produced an experience and the trail reaches information that was supplied, instead of running on without an origin.

A complete mathematical history need not be generated backward. For example, $x_n=n$ satisfies both $x_{(n+1)}=x_n+1$ and $x_n=x_{(n+1)}-1$ for natural-number indices. These relations hold without a last value at infinity. The block objection says such a whole history is enough for experience, with no production at all. A

list of relations that holds in both directions says which states sit where, and nothing in it produces anything. Experience is the producing: each state of a running mind produces the next, and a list of states, like a recording, produces nothing.

Memory alone does not settle it. Recording can be logically reversible, able to be undone because no information is lost (Bennett, 1973). In symbols, let x be a source bit and r a memory bit; the transformation (x,r)->(x,r XOR x) is its own inverse, so starting with r=0 records x and applying it again restores the blank memory. With an independent uniformly distributed initial memory bit, the resulting memory carries no statistical information about x. The useful record depends on the supplied condition, not only the reversible rule.

Appendix B. The scope of a universal reversal penalty

A code can be built that charges the same for a history and its reversal. Stated precisely, let U be an optimal universal prefix-free machine for finite strings and R a fixed computable involution, a reversal that is its own inverse. Define V(0p)=U(p) and V(1p)=R(U(p)). V is prefix-free and universal. Flipping the leading bit pairs every V-program for h with an equal-length program for R(h). Thus K_V(h)=K_V(R(h)).

Both orientations share the interpreter and pay for the selector, and V obtains each reversed string by running a program for the original and reversing its output. So no penalty for finite reversal holds across all universal codes, and the argument here needs none: the asymmetry of Sections 2 and 4 lies in what produces what.

References

Albert, D. Z. (2000). Time and Chance. Harvard University Press.

Barbour, J., Koslowski, T., and Mercati, F. (2014). Identification of a gravitational arrow of time. Physical Review Letters 113, 181101. https://doi.org/10.1103/PhysRevLett.113.181101

Bennett, C. H. (1973). Logical reversibility of computation. IBM Journal of Research and Development 17, 525-532. https://doi.org/10.1147/rd.176.0525

Berlekamp, E. R., Conway, J. H., and Guy, R. K. (1982). Winning Ways for Your Mathematical Plays, Volume 2. Academic Press.

Borde, A., Guth, A. H., and Vilenkin, A. (2003). Inflationary spacetimes are incomplete in past directions. Physical Review Letters 90, 151301. https://doi.org/10.1103/PhysRevLett.90.151301

Crooks, G. E. (1999). Entropy production fluctuation theorem and the nonequilibrium work relation for free energy differences. Physical Review E 60, 2721-2726. https://doi.org/10.1103/PhysRevE.60.2721

Feynman, R. P. (1948). Space-time approach to non-relativistic quantum mechanics. Reviews of Modern Physics 20, 367-387. https://doi.org/10.1103/RevModPhys.20.367

Gardner, M. (1970). Mathematical games: The fantastic combinations of John Conway's new solitaire game "life". Scientific American 223(4), 120-123.

Grunwald, P. D., and Vitanyi, P. M. B. (2008). Algorithmic information theory. In Philosophy of Information. https://arxiv.org/abs/0809.2754

Heinrich, T., Knopp, B., and Pas, H. (2015). Entropy, biological evolution and the psychological arrow of time. arXiv:1401.3734v2. https://arxiv.org/abs/1401.3734

Janzing, D., Chaves, R., and Scholkopf, B. (2016). Algorithmic independence of initial condition and dynamical law in thermodynamics and causal inference. New Journal of Physics 18, 093052. https://doi.org/10.1088/1367-2630/18/9/093052

Landauer, R. (1961). Irreversibility and heat generation in the computing process. IBM Journal of Research and Development 5, 183-191. https://doi.org/10.1147/rd.53.0183

Mlodinow, L., and Brun, T. A. (2014). Relation between the psychological and thermodynamic arrows of time. Physical Review E 89, 052102. https://doi.org/10.1103/PhysRevE.89.052102

National Institute of Standards and Technology (NIST). 2022 CODATA recommended values: Planck time. https://physics.nist.gov/cuu/Constants/Table/allascii.txt

Penrose, R. (2004). The Road to Reality. Jonathan Cape.

Planck Collaboration (2020). Planck 2018 results. I. Overview and the cosmological legacy of Planck. *Astronomy & Astrophysics*, 641, A1.

Price, H. (1996). Time's Arrow and Archimedes' Point. Oxford University Press.

Russell, B. (1903). The Principles of Mathematics. Cambridge University Press.

Russell, B. (1913). On the notion of cause. Proceedings of the Aristotelian Society 13, 1-26.

Solomonoff, R. J. (1964). A formal theory of inductive inference. Information and Control 7, 1-22 and 224-254.

Whitehead, A. N. (1978). Process and Reality, corrected edition, ed. D. R. Griffin and D. W. Sherburne. Free Press. First published 1929.